

Zihan Liang

Undergraduate AI Researcher at Xi'an Jiaotong-Liverpool University

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Research Profile

Undergraduate researcher in artificial intelligence developing reproducible machine-learning methods for mixed-frequency time-series forecasting and sequential social-media risk assessment. Experience spans Transformer and attention architectures, NLP sequence modelling, multi-horizon evaluation, imbalanced classification, and interpretability. Research dissemination includes team research presented at IBEC 2026 and a first-authored poster presented at ICSC 2025.

Education

Xi'an Jiaotong-Liverpool University (XJTLU)	Sep 2024 – Jul 2028, expected
BEng Artificial Intelligence (Intelligent Systems)	Suzhou, China
Dual Degree: BSc (Hons), University of Liverpool, expected	Liverpool, UK
– Academic standing: Completed Stage 2 with a Stage 2 weighted average of 78/100 and an overall weighted average of 74/100.	
– Selected marks: Python for AI 92, Maths for Machine Learning 86, Engineering Mathematics I 82, AI Fundamentals and Ethics 81, and Data Structures and Algorithms 78.	

Research Outputs

- Q. Luo, R. Xu, **Z. Liang**, Z. Hou, and R. Zhao. *Do Macroeconomic and Geopolitical Risk Factors Improve FX Volatility Forecasts? A Mixed-Frequency Transformer Approach*. SSRN preprint, 2025. Preprint | DOI.
- **Z. Liang**, R. Xu, and R. Zhao. *Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media*. Conference poster presented at ICSC 2025. Code | Poster.

Research Experience and Selected Projects

Mixed-Frequency Transformer for USD/CNH Volatility Forecasting	Aug 2025 – Jan 2026
Model Developer, XJTLU	Suzhou, China
– Implemented temporal encoding for mixed-frequency observations and positional encoding for Transformer inputs.	
– Proposed cross-attention using daily technical representations as queries and lower-frequency macroeconomic and geopolitical streams as keys and values.	
– Built multi-horizon forecasting heads and prepared pooled and horizon-specific evaluation tables for 1-, 5-, and 10-day horizons.	
– Produced predicted-versus-actual plots and Integrated Gradients heatmaps to compare model variants and analyze the 2024 U.S. election period.	
– The team presented the research at IBEC 2026.	

Sequential Suicide-Risk Prediction from Social Media	Nov 2025
Lead Student Researcher, First Author of ICSC Poster, XJTLU	Suzhou, China
– Built the RSD-15K pipeline from data acquisition and exploratory analysis through data cleaning, training, and user-level evaluation.	
– Implemented emoji and special-token normalization and head-tail retention for long user histories.	
– Developed a hierarchical architecture combining Twitter-RoBERTa encoding, Bi-GRU sequence modelling, attention, and a user-level classifier.	
– Applied weighted focal loss to address imbalance across risk categories.	
– Achieved weighted F1 0.46 versus 0.43 for a RoBERTa baseline under five-fold user-level evaluation.	
– Prepared the public code and poster repositories and presented the work at ICSC 2025.	

The Fairness Paradox: A Stratified Hybrid-Merit Model	Jan 2026
Team Member, Mathematical Contest in Modeling	Suzhou, China

- The team developed a Softmax-based inverse-inference heuristic to reconstruct latent audience preferences from placements and normalized judges’ scores.
- The team ran counterfactual comparisons of rank- and percentage-based aggregation across competition seasons.
- The analysis compared each aggregation rule with judge-only rankings and examined polarized popularity cases.
- The team applied Kruskal–Wallis tests to study how contestant attributes related to competition longevity.
- The proposed Stratified Hybrid-Merit Model combined percentage-based scoring, a judge-save safeguard, and demographic stratification.
- Named member of a 2026 MCM Finalist team.

SkillNet: Graph-Guided Coordination for Enterprise AI Agents

Jul 2026

Scale Experiment and Data Analysis Lead, SJTU AI Action Summer School

Shanghai, China

- Designed and executed A/B/C experiments comparing flat, department-grouped, and graph-structured catalogues across 46 atomic skills and 21 Gold Tasks.
- Built candidate catalogues and audited experiment configurations and run artifacts for consistent comparison.
- Analyzed Skill selection, department coverage, required ordering, and constraint compliance under explicit and fuzzy-language prompts.
- The graph-structured catalogue recorded 89.13% Skill F1 and 98.10% required-order accuracy in the fuzzy-language experiment. Code.
- Produced comparison tables and result visualizations for the team’s analysis and Demo Day presentation.
- Completed the July 27–31 SJTU summer program and its final team-project assessment.

Grouped NIPT Timing Modelling and Fetal-Abnormality Classification

Sep 2025

Statistical Modeling Contributor, CUMCM 2025

Jiangsu, China

- Studied grouped NIPT testing-time selection and fetal-abnormality classification for CUMCM 2025 Problem C.
- Contributed GLM and GAM analysis of fetal-fraction relationships with BMI and gestational age.
- Developed BMI groupings and a testing-time optimization framework for group-specific recommendations.
- Contributed logistic classification for T21, T18, and T13 using ROC analysis and Youden threshold selection.
- The team documented modelling assumptions, result interpretation, and limitations in the final report.
- The team received Second Prize in the Jiangsu Division, Undergraduate Group.

Au₂₀ Scientific Visualization Support

Oct 2025

Visualization and Reporting Contributor, XJTLU

Suzhou, China

- Built Python utilities for Au₂₀ atomic-cluster structures and bond diagrams. Code.
- Generated atomic-structure and bond-graph figures for the technical report.
- Rendered animation frames in Python and assembled them into LaTeX/PDF presentation sequences.
- Contributed final reporting graphics and presentation assets without claiming authorship of the underlying prediction model.

Selected Honors and Awards

Finalist	2026
Mathematical Contest in Modeling	The Fairness Paradox: A Stratified Hybrid-Merit Model
First Prize for Social Impact	2026
AI Action Summer School, Shanghai Jiao Tong University	SkillNet: Organizing Skills at Scale
Jiangsu Provincial Second Prize	2025
Contemporary Undergraduate Mathematical Contest in Modeling	Grouped NIPT Timing Modelling
Grand Prize, Top 1 out of 50+	2025
XJTLU Research-led Learning Symposium	Au ₂₀ Delta-Learning

Jiangsu Provincial Third Prize	2025
“Keyou Cup” University Intelligent Robot Creativity Competition	Home Repair Assistant
Excellent Poster Award	2025
Summer Undergraduate Research Fellowship	XJTLU

Selected Presentations

International Conference on Social Computing	Nov 2025
Poster Presenter	ICSC 2025
– Presented <i>Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media</i>. – Communicated the sequential modelling approach, user-level evaluation protocol, and poster findings.	
XJTLU Research-led Learning Symposium	Nov 2025
Poster and Oral Presenter	XJTLU
– Advanced through poster screening and delivered an oral presentation followed by a Q&A. – Presented hierarchical suicide-risk modelling and received Second Prize for the project.	

Technical Skills

Programming and tools: Python, Git, and LaTeX.
Python ecosystem: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, and Matplotlib.
Machine learning: Transformers, attention mechanisms, mixed-frequency and multi-horizon forecasting, cross-validation, and imbalanced classification.
NLP and data methods: RoBERTa encoding, Bi-GRU sequence modelling, statistical modelling, and scientific visualization.
Language of Instruction: Degree programme taught entirely in English. Official English Medium of Instruction (EMI) letter available.